Yanjin Li

Self-taught AI Engineer & Researcher — research · systems · applied ML · product

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Self-taught AI builder — learning systematically from papers and open-source code since early 2026, I designed, built and evaluated **11 projects in three months**, spanning training-method research, multi-device AI systems, applied decision systems and shipped products.

ESCP (Grande École, Management & Finance) · prior internships: **BNP Paribas** · **HSBC** · **TikTok** · **DiDi**

SELECTED PROJECTS

Each title links to its full technical write-up at yanjin-ai.github.io/portfolio.

Zero Gradient ↗ — training a large language model without backpropagation, and mapping where that approach hits its limits
RESEARCH

- Asked a concrete question — can local learning rules replace end-to-end gradients on a large model under a hard single-GPU budget — and designed the work to find *where it breaks*, not just whether it runs.
- Built a 4-billion-parameter content-routed mixture-of-experts model with layer-local losses and frozen attention, then a staged series of controls to separate cause from correlation.
- The result I value most is the method: I can design honest ablations and locate a limitation precisely — here, the ceiling turned out to be the architecture, not the missing gradients. Written up as a short paper.

`PyTorch (autograd off) · MoE 950 experts/layer · layer-local learning · EMA routing · WikiText-103 · Kaggle T4`

Gemma4all ↗ — a cross-device assistant where a task can run and hand off across phone, desktop and cloud SYSTEMS

- Started from the question I actually cared about — what's the right trust model for an assistant that can touch your files — and let that drive the whole architecture rather than the demo.
- Built it schema-first and event-sourced: one spec generates the types across four languages, every state change is an immutable event, dangerous tools fail closed behind human approval, and inference runs on-device.
- What I took away: designing the contract — and what should happen when things go wrong — *first* is what makes a system you can actually trust and keep extending.

`TypeScript · Python · Swift · Kotlin · JSON Schema 2020-12 · LiteRT-LM / Metal · SQLite · event sourcing · Bonjour`

TriageGeist ↗ — an emergency-triage decision-support stack built around what a clinician can actually rely on APPLIED ML

- The first thing I did was interrogate the labels — and found the apparent signal was largely a lookup table, so optimizing the score would have shipped something unsafe.
- So I built the honest model on physiology instead, and wrapped it in what a real decision needs: calibrated confidence, a principled way to defer to a human, and a fairness check across subgroups.
- The lesson I kept: in a decision system, knowing *when not to trust the model* matters more than the headline metric.

`LightGBM (multi-task) · isotonic calibration · conformal prediction (APS) · bootstrap fairness · shared transform()`

NeuroGolf (ARC-AGI) ↗ — compiling reasoning tasks into the smallest possible neural circuits RESEARCH

- The insight that unlocked it: this isn't a learning problem — each task's true rule is shipped as a generator, so the job is to *compile* known rules into minimal networks, not to train.
- Reverse-engineered the scoring into a cost model, then built the tooling to construct and validate circuits against freshly generated data so nothing overfits the public samples.
- What I took from it: the difference between fitting something that passes the samples and implementing the rule that produces them — only the second holds up. Written up as a reusable methodology note.

`ONNX / onnxruntime · program synthesis · scorer reverse-engineering · integer-weight circuits · LP / torch fitting`

AI Showrunner ↗ — turning one prompt into a short film through a pipeline with a critic in the loop PRODUCT

- The bet: treat generation like a production pipeline with feedback, not one big prompt.
- Built script → storyboard → per-shot generation → a vision-model reviewer that scores each shot and drives targeted retries, with characters kept consistent across shots and every stage a replayable artifact.
- What I learned: a loop with an evaluator beats one-shot prompting for anything that has to stay coherent end to end.

`FastAPI · asyncio · Pydantic · Qwen-Max planner · Qwen-VL critic · Wan I2V/T2V · ffmpeg`

Nemotron Post-training ↗ — a reproducible post-training pipeline for NVIDIA Nemotron-3-Nano-30B — a 30B MoE + Mamba–Transformer hybrid reasoning model RESEARCH

- Set out to build the whole post-training path cleanly and reproducibly — teacher, data, supervised fine-tuning, reward shaping — for a 30B hybrid model on a single GPU.
- Got the pipeline and a strong rule-based teacher working; when the fine-tuned model decoded poorly despite a healthy training loss, I traced it to a teacher-forcing / generation mismatch and designed the fix.
- The habit that stuck: validate the full train→generate loop on a few samples before committing to a long run.

`Transformers · PEFT/LoRA · Unsloth · TRL · 4-bit quantization · GRPO reward shaping`

Also built: Offline French Companion (in-browser LLM tutor — WebLLM + Whisper + OCR, fully offline) · Orbit Wars (multi-agent evaluation methodology) · Maze Crawler (strategy agent with CI) · VoiceInput (pure-Swift macOS speech input) · Atelier Post (shipped art-to-Instagram tool, web/PWA/iOS — live).

PROFESSIONAL EXPERIENCE

BNP Paribas — Global Corporate Banking Intern — Paris · Mar – Aug 2025

- Structured tailored cross-border transactional and financing solutions for multinational clients; built and presented pitchbooks combining market analysis with pricing and cross-border liquidity.
- Helped land internal AI tooling — built RAG knowledge bases on Open WebUI and Dify to improve client-response efficiency (book incl. CATL, BYD, Alipay, Midea, TCL, JD.com).

HSBC — Global Markets, Liquidity & Investment Intern — Paris · Jun – Nov 2024

- Monitored Fed/ECB/BoE rates and money markets to optimize short-term EMEA portfolios under LCR/NSFR; built automated VBA/Excel tools tracking rate moves and pricing gaps in real time.
- Developed dynamic pricing models for wholesale deposits; contributed to the ALCM committee (gap-risk analysis, funding-cost optimization).

TikTok — Growth Strategy Intern — Beijing · Feb – May 2023

- Contributed to 10+ strategic projects, growing SaaS/AI market share by 23%; ran SQL/Excel analytics for a 20+ member team and lifted user conversion by 13% in one month.

Uber China · DiDi — Product Management & User Operations Intern — Beijing · Aug 2022 – Jan 2023

- Managed 3 business-operations projects; analyzed large datasets in Python, SQL and VBA to build predictive models and data-driven user-interaction strategies.

EDUCATION

ESCP Business School (Paris & London) — Master in Management, Finance — 2023 – 2026

Beijing International Studies University — BA Language & Culture (French, Japanese); minor New Media (CUC); self-taught Psychology (Yale/Coursera)

2019 – 2023

SKILLS

Programming: Python · TypeScript/Node · Swift · Kotlin · JavaScript · SQL · VBA

AI / ML: PyTorch · LoRA/PEFT · MoE · calibration · conformal prediction · LiteRT-LM · ONNX · WebLLM · Whisper · LLM/agent orchestration · schema-first & event-sourced systems

Spoken languages: Chinese (native) · English (fluent) · French (fluent) · Japanese (working)

Strengths: High agency and a self-taught, zero-to-one builder — brave in exploration, highly sensitive and perceptive, reflective and resilient. An avid reader who loves observing the world, with a painter's eye for design; I choose the things I love and strive to do them best.